Yuki Takaha

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Link

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Research interest

Statistical mechanics, nonequilibrium physics, disordered systems, amorphous solids and glasses, self-organized criticality, percolation, active matter, rheology

Work

- **Assistant Professor, Institute of Industrial Science, The University of Tokyo** *April 2026 - present*
  The group of Prof. Akira Furukawa Tokyo, Japan

Education

- **Doctor of Philosophy in the field of Multidisciplinary Sciences** *April 2023 - March 2026*
  Department of Basic Science, Graduate School of Arts and Sciences, The University of Tokyo Tokyo, Japan
  ◦ Advisor: Prof. Atsushi Ikeda

- **Master of Science in the field of Multidisciplinary Sciences** *April 2021 - March 2023*
  Department of Basic Science, Graduate School of Arts and Sciences, The University of Tokyo Tokyo, Japan
  ◦ Advisor: Prof. Atsushi Ikeda

- **Bachelor of Science** *April 2017 - March 2021*
  Department of Physics, Faculty of Science, The University of Tokyo Tokyo, Japan

Papers

Preprint

2. **Y. Takaha**, and A. Ikeda
   "Aging and protocol dependence of thermal activation events in structural glasses"
   arXiv:2610.39405

1. H. Kobayashi, **Y. Takaha**, C. Yuhn, Y. Sato, S. Tomita, A. Kawamoto and T. Nomura
   "Automated design of soft-rigid hybrid robots for dynamic locomotion"
   arXiv:2605.29389

Published papers

3. **Y. Takaha**, H. Mizuno, and A. Ikeda
   "Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
   Physical Review E **112**, L043401(2025), arXiv:2409.15775

2. **Y. Takaha**, H. Mizuno, and A. Ikeda
   "Arrhenius temperature dependence of the crystallization time of deeply supercooled liquids"
   Physical Review Research **6**, 013040(2024), arXiv:2303.07817

1. **Y. Takaha** and D. Nishiguchi
   "Quasi-two-dimensional bacterial swimming around pillars: Enhanced trapping efficiency and curvature dependence"
   Physical Review E **107**, 014602(2023), arXiv:2203.16017

Awards, Fellowships, & Grants

- **Student Presentation Award of the Physical Society of Japan** *Apr. 2026*
  JPS 2026 Spring Meeting
- **3rd Poster Prize Award for the Topic 4-Disordered and glassy systems** *Jul. 2025*
  29th International Conference on Statistical Physics
- **JSPS Doctorial Cource Research Fellowship(DC2)** *Apr. 2024 - Mar. 2026*
  Japan Society for the Promotion of Science
- **Student Presentation Award of the Physical Society of Japan** *Oct. 2023*
  JPS 78th Annual Meeting
- **World-leading Innovative Graduate Study Program of Advanced Basic Science Course** *Apr. 2021 - Mar. 2024*
  The University of Tokyo

Presentation

*Presenting author

International conferences and meetings

11. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
The 10th International Discussion Meeting on Relations in Complex Systems, **Invited talk**, Barcelona, Spain, July 2025

10. H. Kobayashi*, **Y. Takaha**, C. Yuhn, Y. Sato, A. Kawamoto, and T. Nomura
"Topology optimization of soft robots with rigid actuators using the material point method and extended position-based dynamics"
18th U.S. National Congress on Computational Mechanics, Oral, Illinois, USA, July 2025

9. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
29th International Conference on Statistical Physics, Poster, Florence, Italy, July 2025

8. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
Soft and Liquid Matter Physics: Past, Present and Future, Poster, Tokyo, Japan, March 2025

7. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
9th Workshop on Physics between Ecole Normale Superieure and University of Tokyo, Oral, Tokyo, Japan, December 2024

6. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
Seeking Unifying Principles in Equilibrium and Driven Materials, Oral, Capri, Italy, October 2024

5. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality of rearrangements in aging glasses"
Long-term Workshop on Frontiers in Non-equilibrium Physics 2024, Oral, Kyoto, Japan, August 2024

4. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Crystallization dynamics of deeply supercooled liquids"
The 7th International Soft Matter Conference, Poster, Osaka, Japan, September 2023

3. D. Nishiguchi* and **Y. Takaha**
"How dimension matters for bacterial swimming: Enhanced attractions in quasi-two dimensions"
28th International Conference on Statistical Physics, Oral, Tokyo, Japan, August 2023

2. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Crystallization dynamics of deeply supercooled liquids"
28th International Conference on Statistical Physics, Poster, Tokyo, Japan, August 2023

1. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Relaxation and crystallization dynamics of a monodisperse soft sphere glass"
Japan-France joint seminar "Physics of dense and active disordered materials", Poster, Kyoto, Japan, March 2023

Domestic conferences and meetings

11. **Y. Takaha*** and A. Ikeda
"Waiting time dependence of avalanche in structural glasses "
JPS 2026 Spring Meeting, oral, online, March 2026

10. **Y. Takaha*** and A. Ikeda
"Activation processes in the aging dynamics of structural glasses"
JPS 80th Annual Meeting, oral, Hiroshima, September 2025

9. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
The 12th Soft Matter conference, poster, Osaka, December 2024

8. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality emerges by thermal fluctuation in a quiescent glass"
Forefront research in glass and related fields, oral, Chiba, November 2024

7. **Y. Takaha***, J. Si, S. Ishikawa, T. Yasuda, N. Sakumichi, and T. Sakai
"Gelation condition of a dilute polymer solution"
The 11th Soft Matter conference, oral, Tokyo, December 2023

6. **Y. Takaha***, H. Mizuno, and A. Ikeda
"Avalanche criticality of rearrangements in aging dynamics"
JPS 78th Annual Meeting, oral, Iwate, September 2023

5. **Y. Takaha*** and A. Ikeda
"Crystallization dynamics of a monodisperse soft sphere glass"
The 36th discussion meeting on molucular simulation, oral, Tokyo, December 2022

4. **Y. Takaha*** and A. Ikeda
"Relaxation and crystallization dynamics of a monodisperse soft sphere glass"
The 10th Soft Matter conference, poster, Fukuoka, November 2022

3. **Y. Takaha*** and A. Ikeda
"Crystallization dynamics of a monodisperse soft-sphere glass"
JPS 2022 Autumn Meeting Program, oral, Tokyo, September 2022

2. **Y. Takaha*** and D. Nishiguchi
"Quasi-two-dimensional bacterial swimming around pillars: Enhanced trapping efficiency and curvature dependence"
The 5th research meeting of "Information physics of living matters", poster, Hyogo, June 2022

1. **Y. Takaha*** and D. Nishiguchi
"Quasi-two-dimensional bacterial swimming around pillars: Enhanced trapping efficiency and curvature dependence"
JPS 2021 Autumn Meeting Program, oral, online, September 2021